ECE 102 Homework 2

Edi Zhang

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1. **Elementary Signals**

   (a) Consider the signal $x(t)$ shown. Sketch the following:

   i. $y(t) = x(t)(u(t-1) - u(2t-5))$

   Solution:

$$u(t-1) = \begin{cases} 1, & t \geq 1 \\ 0, & t < 1 \end{cases} \tag{1}$$

$$u(2(t-2.5)) = \begin{cases} 1, & t \geq 2.5 \\ 0, & t < 2.5 \end{cases} \tag{2}$$

Thus

$$u(t-1) - u(2(t-2.5)) = \begin{cases} 1, & 1 \leq t < 2.5 \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

and

$$x(t)(u(t-1) - u(2(t-2.5))) = \begin{cases} x(t), & 1 \leq t < 2.5 \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

   ii. $y(t) = \int_{-\infty}^{t} \delta(\tau - 2)x(\tau)d\tau$

   Solution: Note that

$$x(2) = \int_{-\infty}^{\infty} \delta(\tau - 2)x(\tau)d\tau \tag{5}$$

However, because our upper bound is t, the delta function will only contribute if $t \geq 2$. Since $\delta(\tau - 2)$ is zero for all $t \neq 2$

$$y(t) = \begin{cases} x(2), & t \geq 2 \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

Thus $y(t) = x(2)u(t-2)$

iii. $y(t) = x(t) - (2u(t) - \frac{1}{2}r(t) + \frac{1}{2}r(t-4))$

Solution:

Notice that

$$g(t) = \frac{1}{2}r(t) - \frac{1}{2}r(t-4) = \begin{cases} 0, & t < 0 \\ \frac{1}{2}t, & 0 \leq t < 4 \\ 2 & t \geq 4 \end{cases} \tag{7}$$

and

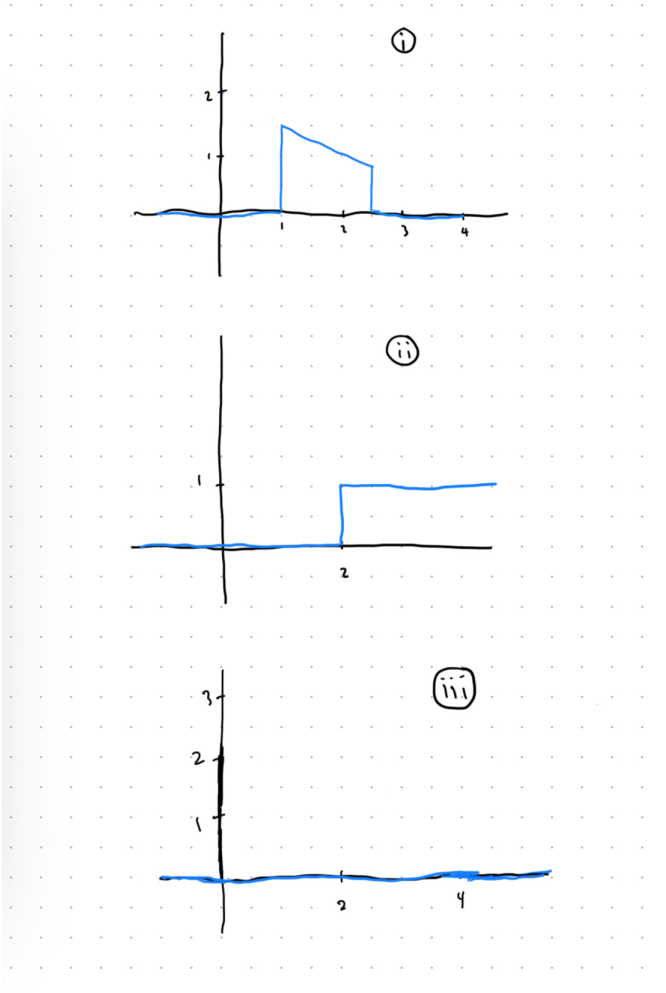
$$2u(t) = \begin{cases} 2, & t \geq 0 \\ 0, & t < 0 \end{cases} \tag{8}$$

Thus

$$2u(t) - g(t) = \begin{cases} 0, & t < 0 \\ 2 - \frac{1}{2}t & 0 \leq t < 4 \\ 0, & t \geq 4 \end{cases} \tag{9}$$

Hence

$$y(t) = x(t) - (2u(t) - g(t)) = \begin{cases} x(t), & t < 0 \\ x(t) - 2 + \frac{1}{2}t, & 0 \leq t < 4 \\ x(t), & t \geq 4 \end{cases} \tag{10}$$



(b) Evaluate these integrals:

i. $\int_{-\infty}^{\infty} f(t+1)\delta(t+1)dt$

Solution:

The spike occurs at $t = -1$. Thus

$$\int_{-\infty}^{\infty} f(t+1)\delta(t+1)dt = \int_{-\infty}^{\infty} f(-1+1)\delta(t+1)dt \tag{11}$$

$$= f(0)\int_{-\infty}^{\infty} \delta(t+1)dt \tag{12}$$

$$= f(0) \tag{13}$$

ii. $\int_{t}^{\infty} e^{-2\tau}u(\tau-1)d\tau$

Solution:

Since

$$u(\tau-1) = \begin{cases} 0, \tau < 1 \\ 1, \tau \geq 1 \end{cases} \tag{14}$$

Therefore

$$\int_t^\infty e^{-2\tau}u(\tau-1)d\tau = \int_t^{1-} e^{-2\tau}u(\tau-1)d\tau + \int_1^\infty e^{-2\tau}u(\tau-1)d\tau \tag{15}$$

Case 1: $t < 1$.

$$\begin{aligned}\int_t^\infty e^{-2\tau}u(\tau-1)d\tau &= 0 + \int_1^\infty e^{-2\tau}u(\tau-1)d\tau && (16)\\ &= \int_1^\infty e^{-2\tau}d\tau && (17)\\ &= \left[-\frac{1}{2}e^{-2\tau}\right]_1^\infty && (18)\\ &= \frac{1}{2}e^{-2} && (19)\end{aligned}$$

Case 1: $t \geq 1$

$$\begin{aligned}\int_t^\infty e^{-2\tau}u(\tau-1)d\tau &= \left[-\frac{1}{2}e^{-2\tau}\right]_t^\infty && (20)\\ &= \frac{1}{2}e^{-2t} && (21)\end{aligned}$$

iii. $\int_0^\infty f(t)(\delta(t-1)+\delta(t+1))dt$

Solution:

$$\begin{aligned}\int_0^\infty f(t)(\delta(t-1)+\delta(t+1))dt &= \int_0^\infty f(t)\delta(t-1)dt + \int_0^\infty f(t)\delta(t+1))dt && (22)\\ &= f(1) && (23)\end{aligned}$$

(c) Let b be a positive constant. Show the following property for the delta function: Hint (what function is "delta-like"

$$\delta(bt) = \frac{1}{b}\delta(t) \tag{24}$$

Solution:

Recall that $\delta(t) = \lim_{\Delta\to 0} \mathrm{rect}_\Delta(t)$. Thus

$$\mathrm{rect}_\Delta(bt) = \begin{cases} \frac{1}{\Delta}, & |t| \leq \frac{\Delta}{2b} \\ 0, & \text{otherwise} \end{cases} \tag{25}$$

Let $\Delta' = \frac{\Delta}{b}$. Then

$$\text{rect}_{\Delta'}(bt) = \begin{cases} \frac{1}{b\Delta'}, & |t| \leq \frac{\Delta'}{2} \\ 0, & \text{otherwise} \end{cases} \tag{26}$$

$$= \frac{1}{b}\text{rect}_{\Delta'}(t) \tag{27}$$

$$\tag{28}$$

Next, as $\Delta \to 0$, $\Delta' \to 0$. Then

$$\delta(bt) = \lim_{\Delta \to 0} \text{rect}_{\Delta}(bt) \tag{29}$$

$$= \lim_{\Delta' \to 0} \text{rect}_{\Delta'}(bt) \quad \text{(Change of Variables)} \tag{30}$$

$$= \lim_{\Delta' \to 0} \frac{1}{b}\text{rect}_{\Delta'}(t) \tag{31}$$

$$= \frac{1}{b}\delta(t) \tag{32}$$

2. **Expression for signals**

   (a) Write the following signals as a combination (sums or products) of unit triangles $\Delta(t)$ and unit rectangles $\text{rect}(t)$.

   i. $y(t) = 2\Delta(\frac{1}{2}(t-2)) - \Delta(t-3)$

   ii. $y(t) = \frac{3}{2}\Delta(\frac{3}{2}t) + \text{rect}(t+\frac{1}{2}) + \text{rect}(t-\frac{1}{2}) - \frac{1}{2}\Delta(2t)$

   iii. $y(t) = 2\Delta(t+1) + 2\Delta(t-1) + \Delta(2t) - \Delta(2(t+\frac{1}{2}) - \Delta(2(t-\frac{1}{2})$

   (b) Express each of the signals shown below as sums of scaled and time shifted unit-step functions.

   Solution:

   (i) $y(t) = u(t+2) + u(t+1) + u(t+\frac{1}{2}) - u(t+\frac{1}{2}) - u(t-1) - u(t-2)$

   (ii) $y(t) = u(t+2) + u(t+1) - 3u(t-\frac{3}{2}) + u(t-3)$

3. **System Properties**

   (a) A system with input $x(t)$ and output $y(t)$ can be time-invariant, causal or stable. Determine which of these properties hold for each of the following systems. Explain your answer.

   i. $y(t) = |x(t)| + x(2t)$

   Solution:

At time $t = \tau$, we need to find $x(2\tau)$, which is the future of τ since (for $t > 0$), $2\tau > \tau$. Thus the system is not causal.

To test for time-invariance, delay the signal by some amount $t_0 > 0$. Then

$$x_{t_0}(t) = x(t - t_0), \quad x_{t_0}(2t) = x(2(t - t_0)) \tag{33}$$

$$\begin{aligned} y(t - t_0) &= |x(t - t_0)| + x(2t - 2t_0) && (34) \\ &= |x_{t_0}(t)| + x_{t_0}(2t) && (35) \end{aligned}$$

Thus the system is time-invariant

To test stability, suppose $\exists M > 0$ such that $|x(t)| \leq M$ for all t. Then

$$\begin{aligned} |y(t)| &= ||x(t)| + x(2t)| && (36) \\ &\leq |M + M| && (37) \\ &= 2M && (38) \end{aligned}$$

Thus $y(t)$ is bounded as well, and the system is stable.

The system is time-invariant and stable, but not causal

ii. $y(t) = \int_{t-T}^{t+T} x(\lambda)d\lambda$, where T is positive and constant

Solution:

Consider

$$x_{t_0}(\lambda) = x(\lambda - t_0) \tag{39}$$

We want to see if

$$\begin{aligned} y_{t_0}(t) &\stackrel{?}{=} \int_{t-T}^{t+T} x_{t_0}(\lambda)d\lambda && (40) \\ &= \int_{t-T}^{t+T} x(\lambda - t_0)d\lambda && (41) \end{aligned}$$

Substituting $u = \lambda - t_0$,

$$= \int_{t-T-t_0}^{t+T-t_0} x(u)du \tag{42}$$

Comparing to,

$$y(t - t_0) = \int_{t-t_0-T}^{t-t_0+T} x(\lambda)d\lambda \tag{43}$$

These are equal. Thus the system is time-invariant

The system is not causal because $T > 0$, and for any t, calculating $y(t)$ would require finding $x(t+T)$, which is in the future of t.

Suppose there exists $M > 0$ such that $x(t) \leq M$ for all t. Thus

$$y(t) = \int_{t-T}^{t+T} x(\lambda)d\lambda \tag{44}$$
$$\leq \int_{t-T}^{t+T} M d\lambda \tag{45}$$
$$= 2TM \tag{46}$$

Since T is constant, $y(t)$ is bounded and thus the system is stable.

The system is stable and time-invariant, but not causal

iii. $y(t) = (t+1)\int_{-\infty}^{t} x(\lambda)d\lambda$

Solution: Consider $x_{t_0}(t) = x(t-t_0)$ Then

$$(t+1)\int_{-\infty}^{t} x_{t_0}(\lambda)d\lambda = (t+1)\int_{-\infty}^{t} x(\lambda - t_0)d\lambda \tag{47}$$
$$= (t+1)\int_{-\infty}^{t-t_0} x(u)du \tag{48}$$

Compare to

$$y(t-t_0) = (t-t_0+1)\int_{-\infty}^{t-t_0} x(\lambda)d\lambda \tag{49}$$

In general, these are not equal, therefore the system is not time-invariant.

The system is causal because at any t, we only need to calculate $x(\tau)$ for $\tau \leq t$.

Since we have no information on $\lim_{t\to\infty} x(t)$, the system is in general not stable. Consider $x(t) = 1$ for all t. Then $|x(t)| \leq 1$, and

$$y(t) = (t+1)\int_{-\infty}^{t} x(\lambda)d\lambda \tag{50}$$
$$= (t+1)\int_{-\infty}^{t} d\lambda \tag{51}$$
$$= (t+1)(\infty) \tag{52}$$

The system is causal, but not time-invariant nor stable

iv. $y(t) = 1 + x(t)\cos(\omega t)$

Solution:

The system is clearly causal because at $y(t)$, only $x(t)$ is needed.

Substituting $x_{t_0}(t)$ for $x(t)$, we get

$$1 + x(t - t_0)\cos(\omega t) \tag{53}$$

Comparing with

$$\begin{aligned} y(t - t_0) &= 1 + x(t - t_0)\cos(\omega(t - t_0)) && (54) \\ &= 1 + x(t - t_0)\cos(\omega t - \omega t_0)) && (55) \end{aligned}$$

The system is time-invariant if $\cos(\omega t) = \cos(\omega t - \omega t_0)$, for all t_0, which isn't true in general. Thus the system is not time-invariant.
Suppose $|x(t)| \leq M$ for some $M > 0$. We know that $|\cos(\omega t)| \leq 1$ for all t. Therefore

$$\begin{aligned} |y(t)| &= |1 + x(t)\cos(\omega t)| && (56) \\ &\leq 1 + |x(t)||\cos(\omega t)| && (57) \\ &\leq 1 + M && (58) \end{aligned}$$

Thus $y(t)$ is bounded, and the system is stable.
The system is causal and stable, but not time-invariant

v. $y(t) = \frac{1}{1+x^2(t)}$
Solution:
Clearly the system is causal, since $y(t)$ only requires $x(t)$ and no future points.
Substituting $x(t - t_0)$ for $x(t)$, we get

$$\begin{aligned} &\frac{1}{1 + x^2(t - t_0)} && (59) \\ &= y(t - t_0) && (60) \end{aligned}$$

Therefore the system is time-invariant.
Suppose $\exists M > 0$ such that $|x(t)| \leq M$ for all t. Then

$$\begin{aligned} |y(t)| &= \left|\frac{1}{1 + x^2(t)}\right| && (61) \\ &\leq \left|\frac{1}{1}\right| && (62) \\ &= 1 && (63) \end{aligned}$$

Thus $y(t)$ is bounded when $x(t)$ is bounded. In fact, it is worth noting the $y(t)$ is always bounded by 1 regardless of whether of not $x(t)$ is bounded. Therefore the system is stable.

The system is causal, time-invariant, and stable

(b) Consider the following three systems:

$$S_1 : w(t) = x(t/2) \tag{64}$$

$$S_2 : z(t) = \int_{-\infty}^{t} w(\tau)d\tau \tag{65}$$

$$S_3 : y(t) = S_3(z(t)) \tag{66}$$

The three systems are connected as

$$x(t) \to S_1 \to S_2 \to S_3 \to y(t) \tag{67}$$

Choose a third system S_3 such that the overall system is equivalent to the following system:

$$y(t) = \int_{-\infty}^{t-1} x(\tau)d\tau \tag{68}$$

Solution:

We want

$$\int_{-\infty}^{t-1} x(\tau)d\tau = S_3\left(\int_{-\infty}^{t} x(\tau/2)d\tau\right) \tag{69}$$

For the RHS, let $u = \tau/2$. Then $2du = d\tau$. When $\tau = t, u = t/2$. Substituting, we get

$$y(t) = S_3\left(\int_{-\infty}^{t/2} 2x(u)du\right) \tag{70}$$

$$= S_3\left(2\int_{-\infty}^{t/2} x(u)du\right) \tag{71}$$

Therefore, let $S_3 : \frac{1}{2}z(2t-2)$

$$y(t) = \frac{1}{2}\left(2\int_{-\infty}^{(2t-2)/2} x(u)du\right) \tag{72}$$

$$= \int_{-\infty}^{t-1} x(u)du \tag{73}$$

(c) In part (b), you saw an example of three systems connected in series. In general, systems can be interconnected in series or in parallel to form what we call cascaded systems. The figure below shows the difference between a series cascade and a parallel cascade. Note that parts (b) and (c) are unrelated.

i. Show that the series cascade of any two time-invariant systems is also time-invariant.

Solution:

Let $S = S_2 \circ S_1$. We want to show that for any t_0,

$$S(x(t-t_0)) = y(t-t_0) \tag{74}$$

Since S_1 is time-invariant,

$$S_1(x(t-t_0)) = S_1(x)(t-t_0) \tag{75}$$

(LHS is evaluating $x(t-t_0)$ then applying S_1, while RHS is applying S_1 to x then substituting $t-t_0$ for t.)

If we now let $w(t) = S_1(x(t))$, we get

$$S_1(x(t-t_0)) = w(t-t_0) \tag{76}$$

Since S_2 is also time-invariant,

$$S_2(w(t-t_0)) = S_2(w)(t-t_0) \tag{77}$$

Then $y(t) = S_2(w(t)) = S_2(S_1(x(t)))$, and

$$S[x(t-t_0)] = S_2(S_1(x(t-t_0))) = S_2(w(t-t_0)) = y(t-t_0) \tag{78}$$

Thus, S is time-invariant.

ii. Show that the parallel cascade of any two time-invariant systems is also time-invariant.

Solution:

We want to show that $S_1(x(t-t_0)) + S_2(x(t-t_0)) = y(t-t_0)$. From part (i), since S_1 and S_2 are time-invariant,

$$S_1(x(t-t_0)) = S_1(x)(t-t_0) \quad \text{and} \quad S_2(x(t-t_0)) = S_2(x)(t-t_0) \tag{79}$$

Therefore

$$\begin{aligned} S_1(x(t-t_0)) + S_2(x(t-t_0)) &= S_1(x)(t-t_0) + S_2(x)(t-t_0) && (80)\\ &= [S_1(x) + S_2(x)]\,(t-t_0) && (81)\\ &= y(t-t_0) && (82) \end{aligned}$$

iii. (Optional) Can you think of two **time-variant** systems, whose series cascade is

time-invariant? Can you think of two **time-variant** systems, whose parallel cascade is **time-invariant**?

Solution:

Series: Consider $S_1(x(t)) = x(t)e^t$ and $S_2(x(t)) = x(t)e^{-t}$. Then

$$\begin{aligned} S_2(S_1(x(t))) &= S_2(x(t)e^t) && (83)\\ &= x(t)e^t e^{-t} && (84)\\ &= x(t) && (85) \end{aligned}$$

This system is clearly time-invariant.

Parallel: Consider $S_1(x(t)) = x(t) + t$ and $S_2(x(t)) = x(t) - t$ as two time-variant systems. Then

$$\begin{aligned} S_1(x(t)) + S_2(x(t)) &= x(t) + t + x(t) - t && (86)\\ &= 2x(t) && (87) \end{aligned}$$

This system is also clearly time-invariant

4. **Power and energy of complex signals**

(a) Is $x(t) = Ae^{j\omega t} + Be^{-j\omega t}$ a power or energy signal? A and B are both real numbers, not necessarily equal. If it is an energy signal, compute its energy. If it is a power signal, compute its power.

Solution:

Compute the energy:

$$\begin{aligned} E &= \int_{-\infty}^{\infty} \left|Ae^{j\omega t} + Be^{-j\omega t}\right|^2 dt && (88)\\ &= \int_{-\infty}^{\infty} \left|A\cos(\omega t) + jA\sin(\omega t) + B\cos(\omega t) - jB\sin(\omega t)\right|^2 dt && (89)\\ &= \int_{-\infty}^{\infty} \left|(A+B)\cos(\omega t) + j(A-B)\sin(\omega t)\right|^2 dt && (90) \end{aligned}$$

Using the property of squared magnitudes of complex numbers, we get

$$\begin{aligned} &\left|(A+B)\cos(\omega t) + j(A-B)\sin(\omega t)\right|^2 && (91)\\ &= \left((A+B)\cos(\omega t) - j(A-B)\sin(\omega t)\right)\left((A+B)\cos(\omega t) + j(A-B)\sin(\omega t)\right) && (92)\\ &= (A+B)^2\cos^2(\omega t) + (A-B)^2\sin^2(\omega t) && (93) \end{aligned}$$

Substituting back in to the energy function,

$$E = \int_{-\infty}^{\infty} \left[(A+B)^2 \cos^2(\omega t) + (A-B)^2 \sin^2(\omega t)\right] dt \tag{94}$$

Integrating $\cos^2(\omega t)$ and $\sin^2(\omega t)$ over infinity will clearly be infinite, thus the signal is not an energy signal. Consider the power instead

$$\begin{aligned}
P &= \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \left|Ae^{j\omega t} + Be^{-j\omega t}\right|^2 dt && (95)\\
&= \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \left[(A+B)^2 \cos^2(\omega t) + (A-B)^2 \sin^2(\omega t)\right] dt && (96)\\
&= \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} (A+B)^2 \left(\frac{1+\cos(2\omega t)}{2}\right) + (A-B)^2 \left(\frac{1-\cos(2\omega t)}{2}\right) dt && (97)\\
&= \lim_{T\to\infty} \frac{1}{2T} \left[\frac{(A+B)^2}{2} \int_{-T}^{T} (1+\cos(2\omega t))dt + \frac{(A-B)^2}{2} \int_{-T}^{T} (1-\cos(2\omega t))dt\right] && (98)\\
&= \lim_{T\to\infty} \frac{1}{2T} \left[\frac{(A+B)^2}{2} \left[t + \frac{\sin(2\omega t)}{2}\right]_{-T}^{T} + \frac{(A-B)^2}{2} \left[t - \frac{\sin(2\omega t)}{2}\right]_{-T}^{T}\right] && (99)\\
&= \lim_{T\to\infty} \frac{1}{2T} \left[\frac{(A+B)^2}{2} \left[2T + \sin(2\omega T)\right] + \frac{(A-B)^2}{2} \left[2T - \sin(2\omega T)\right]\right] && (100)\\
&= \frac{(A+B)^2}{2} \lim_{T\to\infty} \frac{2T + \sin(2\omega T)}{2T} + \frac{(A-B)^2}{2} \lim_{T\to\infty} \frac{2T - \sin(2\omega T)}{2T} && (101)\\
&= \frac{(A+B)^2}{2} + \frac{(A-B)^2}{2} && (102)\\
&= \frac{1}{2}(2A^2 + 2B^2) && (103)\\
&= A^2 + B^2 && (104)
\end{aligned}$$

(b) Is $x(t) = e^{-(1+jw)t}u(t-1)$ and energy or power signal? Again, if it is an energy signal, compute its energy. If it is a power signal, compute its power.

Solution: Recall that

$$u(t-1) = \begin{cases} 1, & t \geq 1 \\ 0, & t < 1 \end{cases} \tag{105}$$

By inspection, since $e^{-(1+j\omega)t}$ decays and $u(t-1)$ zeroes out values where $e^{-(1+j\omega)t}$ is large,

this is likely to be an energy signal.

$$
\begin{aligned}
E &= \int_{-\infty}^{\infty} \left|e^{-(1+j\omega)t}u(t-1)\right|^2 dt && (106)\\
&= \int_{-\infty}^{1-} \left|e^{-(1+j\omega)t}u(t-1)\right|^2 dt + \int_{1}^{\infty} \left|e^{-(1+j\omega)t}u(t-1)\right|^2 dt && (107)\\
&= \int_{1}^{\infty} \left|e^{-(1+j\omega)t}\right|^2 dt && (108)\\
&= \int_{1}^{\infty} \left|e^{-t}e^{-j\omega t}\right|^2 dt && (109)\\
&= \int_{1}^{\infty} \left|e^{-t}(\cos(\omega t) - j\sin(\omega t))\right|^2 dt && (110)\\
&= \int_{1}^{\infty} \left|e^{-t}\cos(\omega t) - je^{-t}\sin(\omega t))\right|^2 dt && (111)\\
&= \int_{1}^{\infty} e^{-2t}(\cos^2(\omega t) + \sin^2(\omega t))dt && (112)\\
&= \int_{1}^{\infty} e^{-2t}dt && (113)\\
&= \left[-\frac{1}{2}e^{-2t}\right]_1^{\infty} && (114)\\
&= \frac{1}{2e^2} && (115)
\end{aligned}
$$